AgriFlow: Agricultural Market Price Analysis and Forecasting

Spatiotemporal Market Integration and Machine Learning-Based Price Prediction

CSI3004 Data Science Programming
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Problem Statement

- Wholesale Onion prices vary sharply across markets and seasons.
- Markets report on uneven dates, which complicates direct comparison.
- Analysts need both descriptive market evidence and a cautious short-horizon forecast.
- Intended users include market analysts, agricultural researchers and other stakeholders.

1 **Uneven market reports**

2 **Validated historical panel**

3 **Analysis plus next-report forecast**

The predictive task estimates the next reported observation, not an exact next-day or current market price.

Objectives

01

Acquire and clean historical market prices

7,794 validated market-day records

02

Characterize trends and reporting coverage

Existing exploratory dashboard

03

Analyze integration and price transmission

Lead-lag, network and geography modules

04

Develop and compare forecasting models

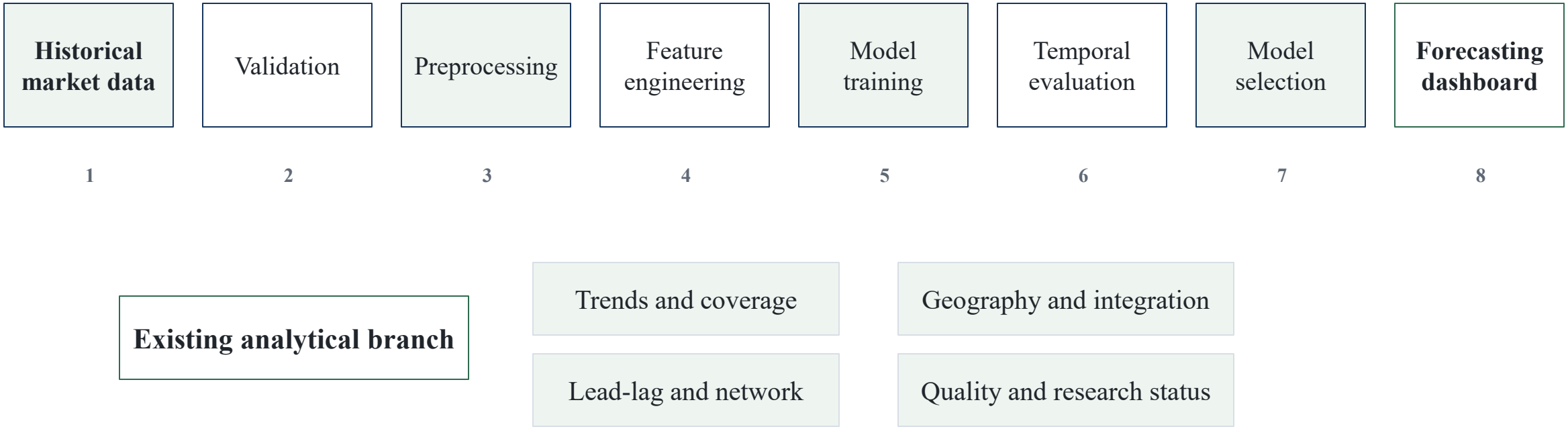
Six candidates on a common temporal cohort

05

Deliver a reproducible analytical dashboard

Offline training, saved inference and evidence

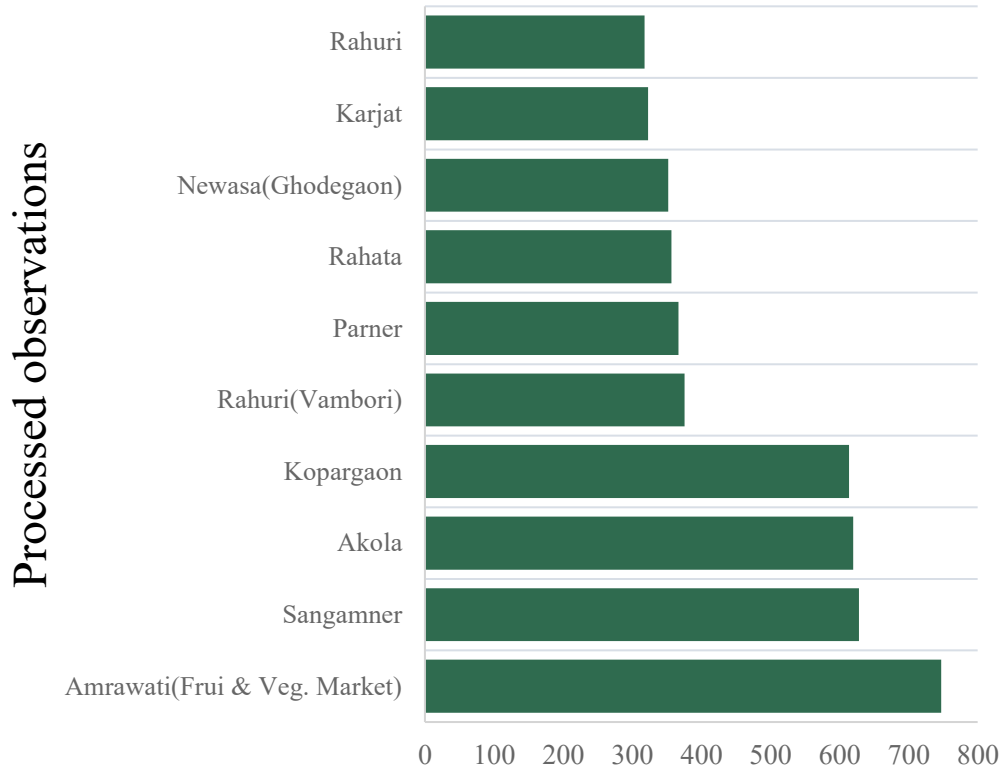
Proposed Solution and Architecture



All normal execution paths use local files. No cloud service, live market API or training inside dashboard callbacks is required.

Dataset Description

Field	Observed scope
Source	CEDA Agri Market Data / AGMARKNET
Commodity	Onion
Coverage	Maharashtra; 4 districts; 25 markets
Rows	8,499 raw; 7,794 processed market-days
Dates	01 Jan 2023 to 30 Oct 2025
Target and unit	modal_price; INR per quintal



Experimental evidence covers Maharashtra only. The India map remains a visualization capability, not a claim of national experimental coverage.

Data Preprocessing and Exploratory Analysis

- Parsed dates and enforced positive numeric modal prices.
- Removed duplicate market-date-commodity targets; none remained.
- Kept missing reports absent rather than filling prices with zero.
- Built market-specific lags and shifted rolling statistics.
- Excluded 322 rows lacking complete 14-report history.



Observed row flow: 8,499 raw checkpoint rows to 7,794 processed market-days to 7,378 modeling rows across 23 eligible markets.

Methodology

Forecast horizon

Next reported market observation

TRAIN	VALIDATE	TEST
through 13 Jan 2025	14 Jan to 07 Jun 2025	08 Jun to 30 Oct 2025
5,199 rows	1,106 rows	1,073 rows

- Global target-date boundaries prevent later targets from entering training.
- Every rolling feature uses a one-observation shift within its market.
- Validation MAE selects the model; validation RMSE resolves ties.
- The held-out test set supports the final comparison only after selection.

Model Implementation

Model	Category	Principal parameters	Advantage	Drawback
M0 Persistence	Baseline	prediction = lag 1	Strong local benchmark	Cannot anticipate turning points
M1 Lag-7	Baseline	prediction = lag 7	Simple longer memory	Ignores recent reports
M2 Ridge	Linear	alpha 1.0; scaled	Stable and interpretable	Linear response
M3 Decision Tree	Tree	depth 8; leaf 5	Nonlinear rules	Overfitting risk
M4 Random Forest	Ensemble	200 trees; depth 12; leaf 3	Robust nonlinear fit	Less transparent
M5 Hist Gradient Boosting	Boosting	150 iterations; rate 0.05; 15 leaves	Efficient nonlinear model	Sensitive to drift

Python 3.12; pandas 2.3.3; NumPy 2.5.3; scikit-learn 1.9.1; deterministic seed 42.

Evaluation Metrics

Metric	Definition	Unit / preferred direction
MAE	mean absolute error	INR/q; lower
MSE	mean squared error	squared INR/q; lower
RMSE	square root of MSE	INR/q; lower
R-squared	1 - SSE/SST	unitless; higher
MAPE	mean absolute percentage error	percent; lower
sMAPE	symmetric percentage error	percent; lower
WAPE	weighted absolute percentage error	percent; lower
Median AE	median absolute error	INR/q; lower
Mean bias	mean(predicted - actual)	INR/q; near zero
Directional accuracy	matching movement signs vs lag 1	percent; higher

MAE = mean(|y - yhat|)

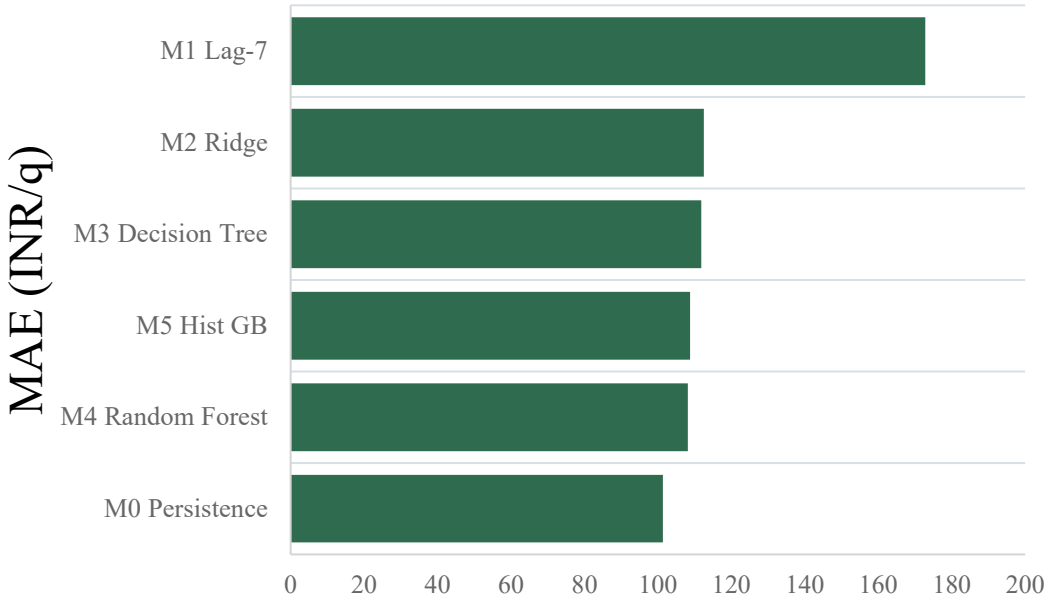
RMSE = sqrt(mean((y - yhat)^2))

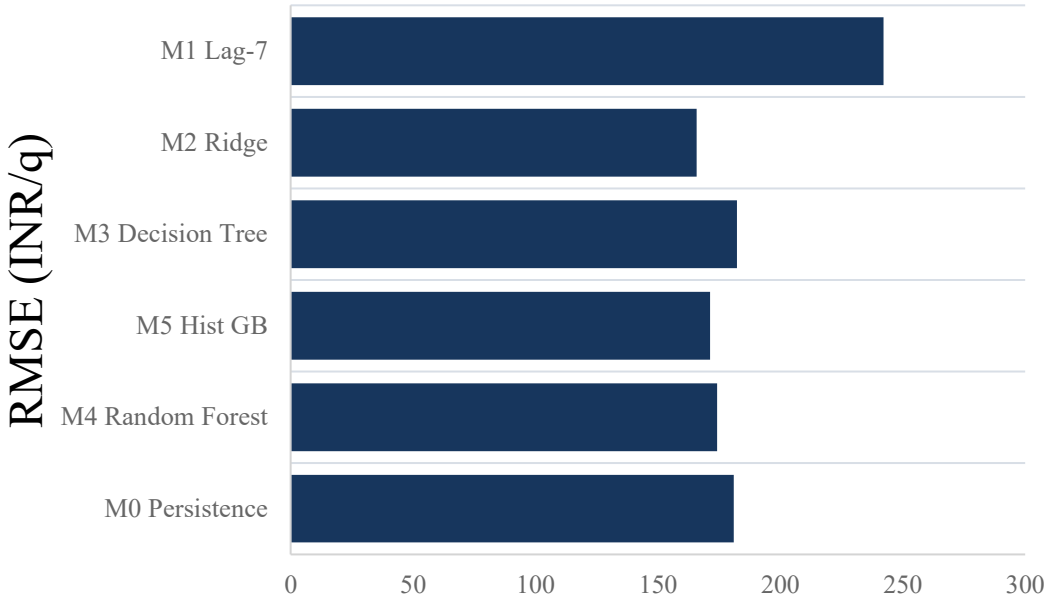
MAPE = 100 mean(|(y - yhat) / y|)

Direction = sign(price - lag 1)

Selection rule
Lowest validation MAE; validation RMSE breaks ties

Model Performance Comparison

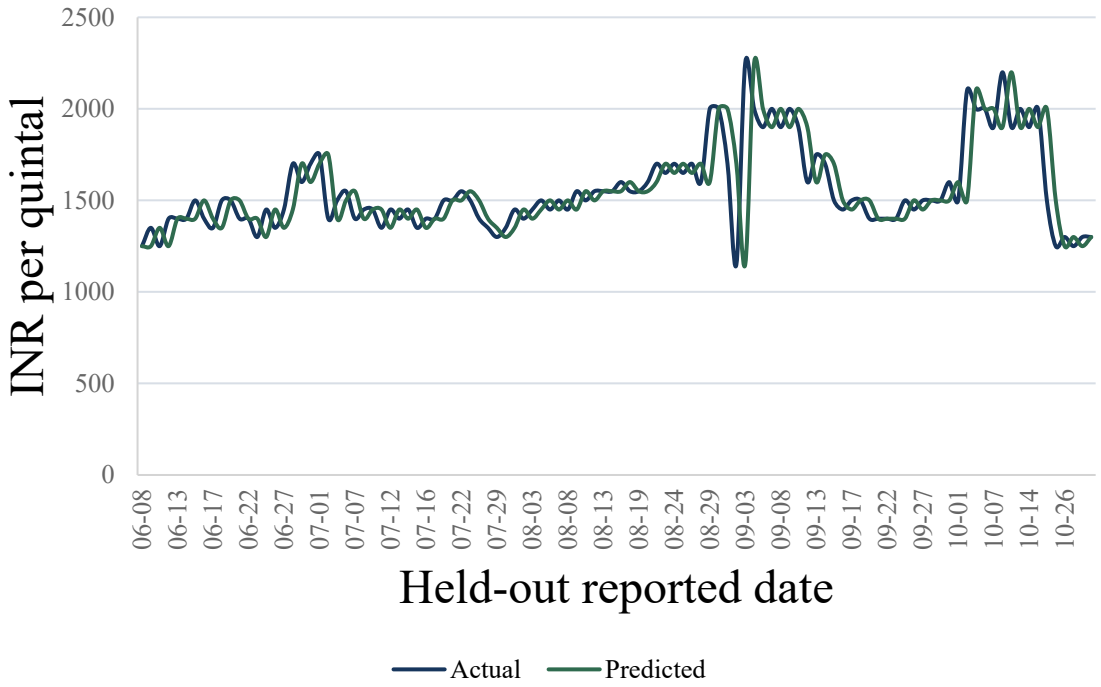


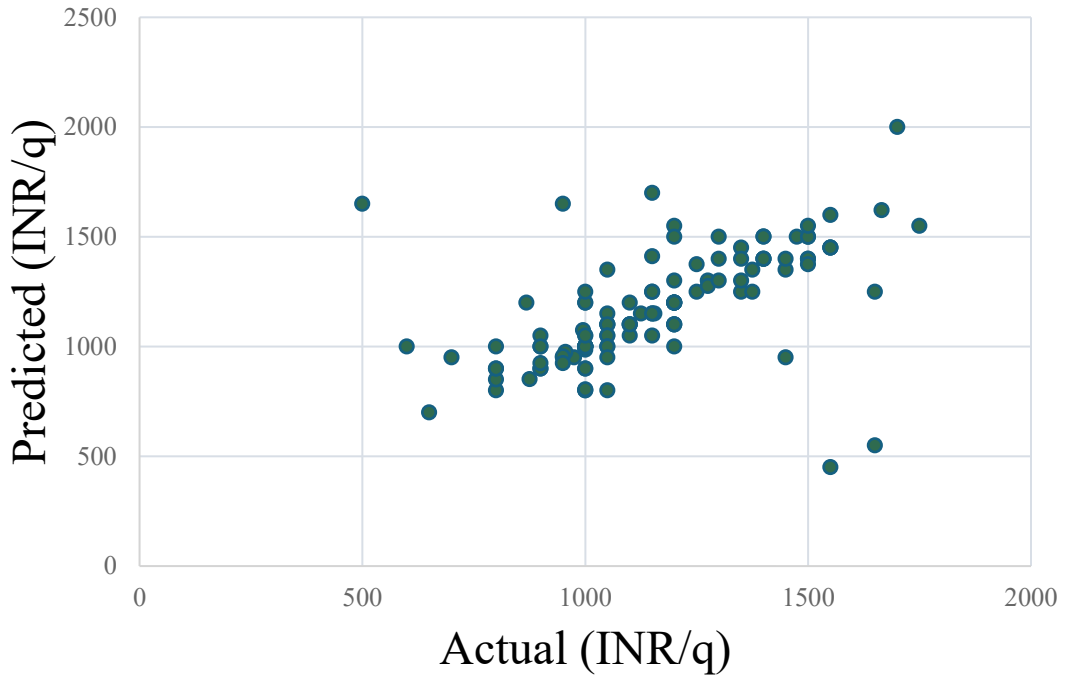


Validation-selected model	M0 Persistence
Validation MAE	128.2 INR/q
Held-out test MAE	101.4 INR/q
Held-out test RMSE	181.0 INR/q
Improvement vs persistence	0.0%

Persistence won the validation rule. No machine-learning candidate improved the selected validation MAE; the baseline therefore remains the preferred model.

Prediction Results and Diagnostics



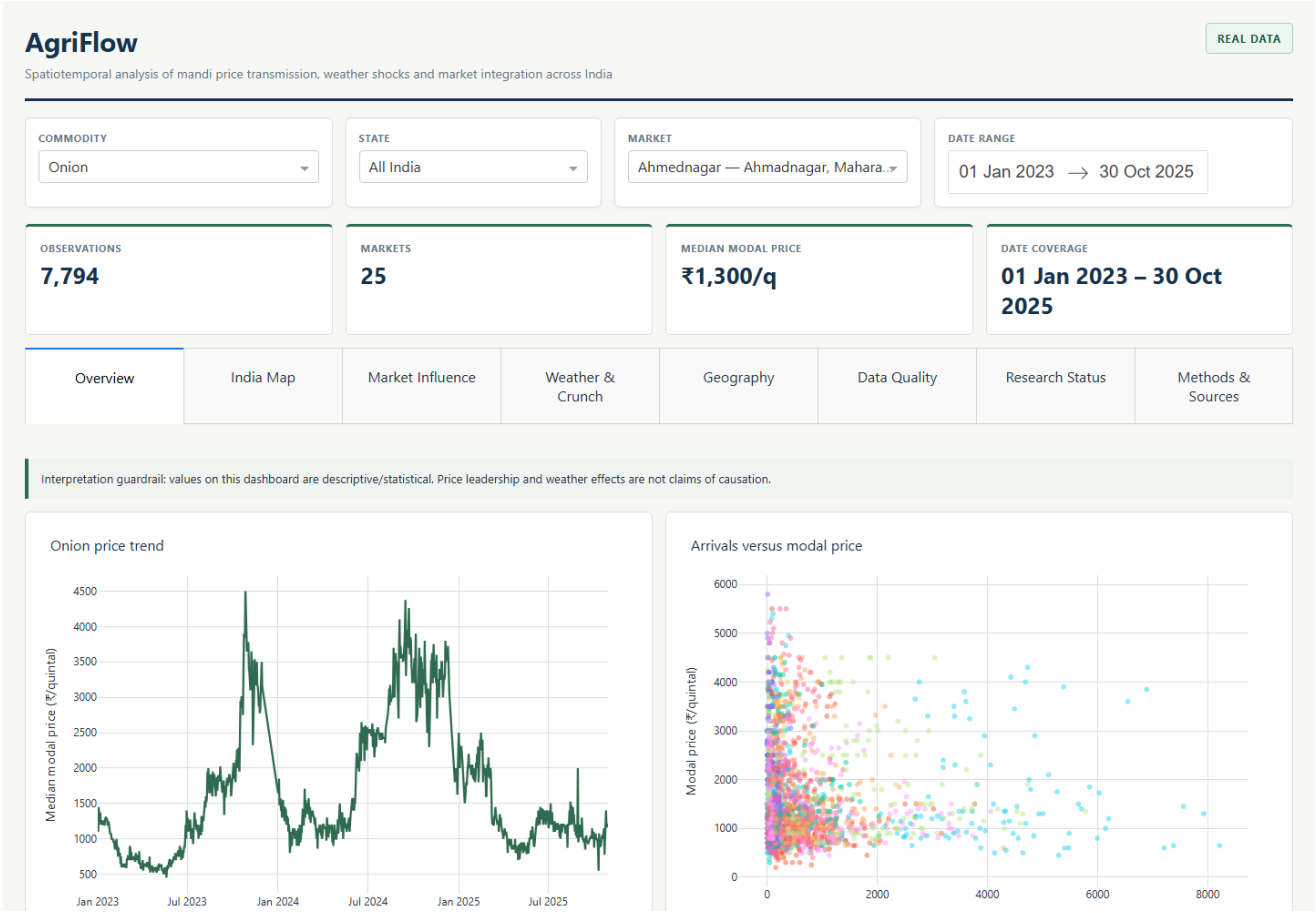


Focus market: Amrawati(Frui & Veg. Market)
Test range: 2025-06-08 to 2025-10-30

Mean bias: -1.2 INR/q
Median absolute error: 50.0 INR/q

Residual convention: predicted price minus actual price. Positive bias indicates overprediction.

Application and Demonstration



Saved forecast output

Market
Ahmednagar

Last observation
2025-10-27 | INR 1150/q

Next-report forecast
INR 1150/q

Model: M0 Persistence
Held-out MAE: 101.4 INR/q

The dashboard also retains Overview, India Map, Market Influence, Weather and Crunch, Geography, Data Quality and Research Status views.

Challenges, Solutions and Drawbacks

Challenge	Implemented response or remaining limitation
Uneven reporting	Define horizon as next reported observation; never fill missing dates as prices.
Restricted acquisition	Train and run offline from traceable local CEDA checkpoints.
Limited geography	State four-district Maharashtra scope; avoid national performance claims.
Temporal leakage	Group by market, shift every rolling feature and split on global target dates.
Overfitting	Use constrained trees, a common cohort and validation-only selection.
Historical staleness	Display the last observation and 2025-10-30 data cutoff.
No exogenous variables	Keep the first model price-only; arrivals units remain uncertain.
Model drawback	Persistence cannot anticipate shocks or turning points, despite winning validation MAE.

Conclusion and Future Scope

What the project achieved

- Preserved the existing market-analysis dashboard.
- Evaluated six forecasting approaches with ten metrics.
- Selected M0 Persistence using validation MAE of 128.2 INR/q.
- Measured held-out MAE of 101.4 INR/q and RMSE of 181.0 INR/q.
- Delivered saved next-report forecasts and reproducible evidence.

Future scope

- Broaden market and commodity coverage.
- Add verified weather and consistently measured arrivals.
- Collect a longer historical series and refresh the data.
- Evaluate fixed calendar horizons separately.
- Revisit tuning only after stronger coverage becomes available.

Measured improvement versus persistence: 0.0%
The baseline won, so no unsupported ML advantage is claimed.

References

- CEDA Agri Market Data documentation: https://api.ceda.ashoka.edu.in/documentation/
- Government of India AGMARKNET / OGD resource: https://data.gov.in/resource/current-daily-price-various-commodities-various-markets-mandi
- AgriFlow repository documentation: README.md; docs/04_DATA_SPECIFICATION.md; docs/05_ANALYTICS_SPECIFICATION.md
- scikit-learn documentation: https://scikit-learn.org/stable/
- pandas documentation: https://pandas.pydata.org/docs/
- NumPy documentation: https://numpy.org/doc/
- Dash documentation: https://dash.plotly.com/
- Plotly Python documentation: https://plotly.com/python/
- Hyndman and Athanasopoulos, Forecasting: Principles and Practice, https://otexts.com/fpp3/

All experimental values in this deck originate from the saved local dataset and generated forecasting artifacts.